

Introduction

This Fast Track gets you started with the Android Software Development Kit (SDK) and helps you understand the ecosystem from a business perspective. You will be introduced to Android as a development platform and will learn to set up your environment for Android application development. We will cover some of the best practices in setting up your workspace in addition to listing out various steps for installation, compilation, testing and deployment.

To create projects and show the various codes and their significance in application development, we'll be using the Android Development Tools (ADT) plug-in for Eclipse IDE. You will get an insight into the workflow of application development, and be able to test-run it on Android Virtual Device (AVD, an emulator) or on a physical device (such as an unlocked Google Nexus S developer phone). This guide also runs you through some basic executable programs, and as is done in any programming tutorial, we start off with creating a 'Hello World!' project and compiling it to run on an emulator.

Once you're well versed with basic workspace components used for application development, you'll be shown how to create layouts and views to produce compelling user interfaces. We'll acquaint you with the basics of coding in Java and XML along with a listing of essential Widget properties. You'll also learn to log in to Android using LogCat. Just like in any computer program, you need to thread your process to create efficient and responsive programs. We'll introduce you to the basics of multi-threading in Android applications. The user interface section will teach you to optimize your layout and check the running process using Hierarchy Viewer. We'll learn to implement the same for a Twitter Status Update application.

Since your application will be deployed on a portable device capable of pin-pointing your location through network-based GPS,